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DISSERTATION: LITERATURE REVIEW

How Do Constructed Emotions Influence Economic Models of Well-being? A Critical Examination

Full name of author:

Jérémie de Rijk

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I declare that this dissertation is my own work, and that where material is obtained from published or unpublished work, this has been fully acknowledged in the references.

Signed: Jérémie de Rijk 10/01/2024

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Abstract

This dissertation investigates the integration of non-rational human behaviour, through the lens of the theory of constructed emotions, into economic models of happiness. By applying this theory to existing models and statistical assumptions, it highlights what existing tools within economics can be applied in the circumstances. This multi-disciplinary approach underscores a broader move within the study of the Economics of happiness to embrace non-rational human nature and how it may provide context and information to traditional viewpoints.

Introduction

1.1 Subjective well-being

In Economics, "utility" is an individual's satisfaction or happiness derived from the purchase or consumption of a good or service. In the last century, Neoclassical economists have mostly inferred an individual's utility from the decisions they make in the market. This is under the premise that an individual's preferences and desires are revealed in their observable market behaviour. Their understanding was that, under a few assumptions, an individual's decisions in the market would always be ones that maximise their expected utility. These assumptions are that; all individuals are rational, fully informed, and always seek to maximise utility (Dolan, Peasgood and White, 2008, p.95).

However, in more recent times, economists and psychologists have become increasingly sceptical that preferences are not a very good guide to understanding people's well-being and happiness (Dolan, Peasgood and White, 2008, p.95). The assumptions required, especially rationality, have been challenged. Numerous studies indicate that individuals often perform poorly when faced with decisions that have objectively correct answers, as they often misunderstand statistical principles and reasoning and exhibit cognitive biases and errors in judgment (Colinsk, 1996, p.670). Furthermore, Easterlin (2003, p. 11179) suggests a fallacy with the revealed preference approach. If one truly believes that the only person who can judge a person's feelings and well-being is concerned, then the belief that an outsider could judge an individual's happiness purely by observing their consumption is fundamentally flawed. One, instead, is led to look at self-reported levels of happiness.

The study of self-reported subjective well-being is part of a general move to incorporate nonrational human behaviour into economics (Adler and Fleurbaey,2016, p.424). Subjective well-being is a multidimensional entity that involves a cognitive component, concerning how we evaluate life satisfaction, and an affective component, related to our positive and negative reactions (Albuquerque et al., 2012, p.448).

In these studies, the participants answer subjective questions based on terms such as 'happiness', 'general satisfaction' and 'subjective wellbeing'. These answers can be categorised using numerical categories, e.g., using 11 numerical categories (scale from 1 to 10) or by using clearly ordered verbal labels, e.g., very happy/happy/somewhat happy/very unhappy (Ferrer-i-carbonell and Frijters, 2004, p. 643).

These results are then modelled with a standard form of:

$$W_{it} = \alpha + \beta X_{it} + \varepsilon_{it}$$

Where W is the reported well-being of an individual I at time t, X is the vector of known variables that may include sociodemographic and socioeconomic characteristics. Unobserved characteristics and measurement errors are included in the error term (Adler and Fleurbaey, 2019, p. 428).

These regressions yield lower R-squared or pseudo-squared values (depending on the estimation process) than economists are used to. Furthermore, in natural environments, reported well-being questions can fluctuate over short periods of time. For example, in a survey done in 2004 where individuals were

interviewed only two weeks apart, it was computed that there was only a correlation of 0.59 across individuals (Kahneman D and Krueger A.B, 2006, p. 7). Reflecting the extent to which emotions and other components of true well-being are driving the results (Adler and Fleurbaey, 2019, p. 428).

1.2 The theory of constructed emotion

As the world of economics looks to incorporate nonrational human behaviour into the study of happiness (Adler and Fleurbaey,2016, p.424), it is important to consider the fields of neuroscience and psychology when formulating models, as they might provide clarity and context to our models and help us understand reasons for some inaccuracies.

The theory of constructed emotion, proposed by Dr Lisa Feldman Barrett, posits that our brains did not evolve for rationality, happiness, or accurate perception, but to efficiently ensure resources for physiological systems (2017, p. 3). Barrett suggests that an individual's emotional experiences are actively constructed by our brains using past experiences and incoming stimuli and that emotions are not universally fixed or static but are complex and dynamic based on individual perceptions and experiences (Barrett, 2017, p.9). "Affect" is seen as the foundation of emotional construct, which are the basic feeling states of pleasure vs displeasure. Building upon affect, cognition is the construction of feelings and emotions based on current situations, past experiences, and cultural understanding (Barret,2017, p.16).

In economic models, it is often assumed that one can separate the two and model the pure cognition behind the effect of socioeconomic factors (Adler and

Fleurbaey, 2016, p.425). However, Barrett collapses the barriers between the cognitive and the affective, emphasising the interplay between the two (Barrett,2017, p. 16).

In this analysis, I aim to explore how the theory of constructed emotions may provide context and a better understanding of different subjective well-being models and how it affects statistical assumptions made within them. This will hopefully create an understanding of the reasons for a range of problems in the research which includes “some contradictory evidence, concerns over the impact on the findings of potentially unobserved variables and the lack of certainty on the direction of causality” (Dolan, Peasgood and white, 2008, p. 94). In the course of this examination, the implications of cognition and affect will be investigated by making statistical inferences about them and their consequences on different frameworks for modelling subjective well-being.

2. Implications of cognition

2.1. Models assuming ordinal comparability

Models assuming ordinal comparability are usually of ordered latent-variable form, in either ordered probit or ordered logit form, which is solved by maximum likelihood or by logistic regression. For instance, Blanchflower and Oswald used both the ordered probit model (2000a) and the ordered logit model (2000b). They are in the general form of:

$$W_{it}^{*} = \beta x_{it} + \varepsilon_{it}$$

$$W_{it} = k \leftrightarrow \lambda_k \leq W_{it}^{*} < \lambda_{k+1}$$

Where W_{it}^{*} is the latent variable, W_{it} is the observed wellbeing value, x_{it} are the socioeconomic factors we are measuring the effect of and ε_{it} is typical idiosyncratic error (Ferrer-i-Carbonell and Frijters, 2004, p.648).

To be regarded as causal, these models require that unobserved factors, ε_{it} or v_i, are either unrelated to observed factors or their relationship is known: $cov(\varepsilon_{it}, x_{it}) = z_{it}^{1}$ and $cov(v_i, x_{it}) = z_{it}^{2}$, with z either 0 or a known function. v_i is a fixed linear individual trait. If this assumption cannot be met this could cause inaccurate estimates (Ferrer-i-carbonell, 2004, p.644).

In a constructed emotions context, this assumption may be concerning and may be the cause for some inaccuracies in these specific models. Dr Barrett (2008) posits that emotions are constructed from a range of physiological operations and are not fixed responses, variation is the norm. Suggesting that differences in the cognitive part of emotional construction, through; past experiences, cultural understandings of happiness, current context, conceptualization of emotions etc.,

is a time invariant characteristic influencing one's fixed linear individual trait or emotional construction v_i . These largely do not fluctuate over the short term.

These time-invariant characteristics, in theory, would be extremely hard to model, as to understand the nature of emotion, one must also model brain systems (Barrett, 2017, p.16). Therefore, their relationship to other observed factors would be largely unknown because collecting data on personal values, cultural background, past experiences, etc., would be a very hard task to undertake.

This creates time-invariant parameter heterogeneity, which is where regression coefficients of the estimating model differ across observations or groups of observations (Zietz,2001, p.264). In this case, these time-invariant characteristics are crucial for creating a backdrop for emotional experiences, from which subjective well-being is assessed. These will systematically influence the observable variables related to wellbeing; $cov(v_i, x_{it}) \neq 0$.

Furthermore, this may create some re-evaluation of what was previously considered idiosyncratic error in models of this form. What was previously attributed to unexplained idiosyncratic variability, may be systematic and related to an individual's constructed emotional experience creating bias in the estimated coefficients.

2.2. Models assuming cardinal comparability

One way of allowing for fixed linear differences is by using a Fixed effect model, which is popular in psychology. They have typically been OLS-estimated cardinal models (Ferrer-i-Carbonell and Frijters, 2004, p.645). For instance, in

economics, Richard Easterlin has modelled subjective well-being using cardinality in various studies (1974; 1995).

To be seen as causal, these models can allow for time-invariant factors, related to initial levels of observed factors, v_i (which in our case would be individuals cognitive construction), but require that time-varying unobserved factors ε_{it} are unrelated to observed factors: $cov(\varepsilon_{it}, x_{it}) = cov(v_i, \Delta x_{it}) = 0$ and $cov(v_i, x_{it}) \neq 0$ (Ferrer-i- Carbonell and Frijters, 2004,p. 644).

These models are in the general form of:

$$W_{it} = \beta x_{it} + v_i + \varepsilon_{it}$$

By taking the first-difference estimator the fixed linear trait v_i would drop out:

$$W_{it} - W_{it-} = \Delta x_{-t}\beta + \Delta\varepsilon_{it}$$

For instance, Gerlach and Stephan (1996) and Korpi (1997) have used a fixed-effects framework on individual-level data, to let linear traits drop out (Ferrer-i-Carbonell and Frijters, 2004, p.646). In the context of the theory of subjective well-being, the unobserved emotional constructions and actions are shaped by one's previous knowledge and experience (cognition), which creates an initial level of observed factors in these models, which will differ across individuals (Barrett, 2017, p.9).

Models in these forms will allow us to model the changes in the variables we are measuring, therefore understanding the effect they have on Well-being.

2.3. Fixed effect ordinal response models

Since the work of Pareto, many economists have been reluctant to assume that Well-being Is cardinal (Dolan and Peasgood, 2008, p.13). Cardinal comparability is the assumption that one can add specific and quantifiable values to subjective experience, e.g., the difference between an answer on happiness of 8 and 9 is the same as the difference between a 4 and a 5 (Ferrer-i-Carbonell and Frijters, 2004, p.644). Well-being or happiness is neither a direct nor verifiable experience, which is constructed only when asked for, being determined by an individual's current mood and memory as well as immediate context (Kahneman D and Krueger A.B, 2006, p.13). As a result, a specific and quantifiable point on a well-being/happiness scale might not represent the same intensity of emotion for different individuals.

In the context of the theory of constructed emotion, the scepticism in cardinality is well-founded. If emotions are constructed and subjective, the intensity of these emotions cannot be measured, or compared. An individual's emotional response to certain socioeconomic factors can vary based on their unique experiences and culture as well as genetics (Barret, 2017, p. 16).

Combining fixed effect and ordinal models helps us solve the issue of time-invariant heterogeneity from differences in cognition and scepticism in cardinality. Two papers which have used this approach are Winkelmann and Winkelmann (1998) and Hamermesh (2001). Their models are in the form of:

$$W_{it}^{*} = \beta x_{it} + v_i + \varepsilon_{it}$$

$$W_{it} = I(GS_{it}^{*} > 0)$$

(Ferrer-i-Carbonell and Frijters, 2004, p.651)

Where W_{it}^{*} is the latent variable and W_{it} is the observed wellbeing value. This is a dichotomous model with fixed effects. This means there are only two categories, which the data needs to be reduced to fit this model. For example, Winkelmann and Winkelmann reduced the (0,10) scale to whether well-being is higher than 7 or not. In their framework, this must be done to accommodate for fixed effects.

2.4. Problems with ordinal comparability

Ferrer-i-Carbonell and Frijters (2004) take it one step further, proposing a fixed effect ordered logit model. In the form of:

$$W_{it}^{*} = \beta x_{it} + v_i + \varepsilon_{it}$$

$$W_{it} = k \leftrightarrow \lambda_k^i \leq W_{it}^{*} < \lambda_{k+1}^{it}$$

Importantly, this model has individual specific thresholds, λ_k^i and ordinal comparability is not assumed.

Ordinal comparability is the assumption that "individuals share a common opinion on what happiness is", where one can make meaningful comparisons between well-being scores, without implying the exact differences between them. This assumption has been supported by supposed evidence. For example, Sandvik et al. (1993) posited that individuals were able to recognise and predict the

satisfaction level of others. In interviews, respondents were shown pictures and videos and were somewhat able to accurately decipher whether they were happy, jealous, etc., even across different cultures. This evidence has helped ordinal comparability become popular in models of Well-being in Economics (Ferrer-i-Carbonell and Frijters, 2004, p.650).

Dr Barrett (2017, p.2) wholeheartedly refuses this claim, refuting the evidence, suggesting it is an old-fashioned view of emotions. New evidence has shown that facial expressions and body movements are not nearly enough to understand what one is feeling. This is especially true when comparing across cultures, where facial expressions are rarely fully translated. If this assumption can no longer be made, more needs to be done to assume ordinal comparability in models. Therefore, a model which does not assume ordinal comparability is one which, in the context of the theory of constructed emotion, provides more accurate estimates.

Furthermore, the use of individually specific thresholds, λ_k^i indicates the differences individuals would cognitively construct their evaluative well-being based on experience, cultural background, and other time-invariant factors. This allows the individual to influence at which point they move from one category to the next. (Barrett, 2017, p. 16).

2.5. Model comparisons

Having looked at different examples of modelling techniques in the study of subjective well-being, using data from the German socio-economic panel, we

can inspect the differences in the values of coefficients depending on the technique used to estimate them. (Ferrer-i-Carbonell and Frijters, 2004, p.653).

	Fixed-eff. ordered logit		Fixed-eff. OLS		Ordered logit	
	Estimate	t-val	Estimate	t-val		
Age					−0.07	10.3
Age × age	−0.0002	0.7	−0.0006	6.5	0.001	12
ln(household income)	0.19	3.8	0.11	4.3	0.47	17.8
Number of children	0.002	0.1	0.01	0.9	−0.06	5.6
Steady partner (1 = yes)	0.08	1.4	0.07	2.4	0.28	11.8
Subjective health	0.37	24.6	0.32	44.1	0.54	88.2
Time-dummies	Yes		Yes		Yes	
Number of individuals	5,801		6,664		7,806	
−log(Likelihood)	9004.3				51004	
R^2			0.09			
Number of cases	25,442		21,104		30,569	

Table 1 - Results on subjective well-being changes in the GSOEP.

The coefficients in ΔW are similar, in both Fixed effect models, even if estimated using different techniques and using different assumptions about the data. This similarity may be due to evidence that claims that respondents try to work out what the interviewer is asking, meaning that they adapt their scale to what they expect the interviewer to expect (Schwarz, 1995).

Contrastingly, compared to the Ordered Logit Model, which cannot be estimated allowing for fixed effects, the coefficients are very different (Ferrer-i-Carbonell and Frijters, 2004, p.653). As expected, T values are lower for Fixed effect models due to a reduction in degrees of freedom.

So far, the inference has been made from the theory of constructed emotions about the time-invariant nature of the cognitive process on the way

individuals evaluate their lives. By using fixed effect models, this time-invariant effect, as well as others not included is mitigated. This allows us to estimate coefficients of socioeconomic factors more accurately in subjective wellbeing models, when in the context of the theory of constructed emotion.

In assessment, it is important to note that while cognitive evaluation remains mostly stable, it slowly evolves as one continues to gain new experiences and learn new things (Barret, 2017, p.9). Therefore, it is imperative to not allow the panel data to have too long a time difference between the beginning and end.

3. Implications of affect

3. 1 Time-variant effect of affect

For causal inferences to be made in all the above models, there must not be any time-varying unobserved factors, ε_{it}, related to observables in an unknown way (Ferrer-i-Carbonell, 2004, p.644). However, in the context of constructed emotions, just as cognition has been inferred to be time-invariant, affect can be inferred to be time-varying. Remember, affect is the foundation of the way individuals construct emotions, which cognition builds upon. It is the very basic part, referring to pleasure to displeasure, which influences our perception and how we interpret information (Barrett, 2017, p.17). affect can change frequently and rapidly; this leads to variability in the affective component of subjective well-being. This introduces time-varying parameter heterogeneity, which is where characteristics of individuals in a population change over time, leading to dynamic differences among them. In this case, the affect would be this dynamic characteristic. The psychological literature demonstrates that affect can bias an individual's thought process and decision-making (Chhaochharia et al., 2019).

For example, Schwarz and Clore (1983) found that subjects primed with positive affect conditions (recall of positive life events and sunny day conditions) reported higher subjective well-being scores than subjects primed with negative affect conditions (recall of negative life events and rainy-day conditions). This is interpreted as evidence that people use affective states as a factor in evaluations of their subjective well-being, meaning it is important to include this in our models. This can lead to misattribution of what is considered idiosyncratic error. What was previously considered idiosyncratic, may be systematic.

3.2 The use of Self-reported affect

For instance, Yardley and Rice (1991, p.104) used Diener and Emmons (1984) self-reported mood scale where, on a scale from "not at all" to "extremely much", respondents were asked the extent to which they felt nine different affective states including depression, happy, pleased, worried, etc. Next, they modelled subjective well-being to mood directly.

This can be applied to a subjective well-being model including socio-economic factors as well as mood measured in economics. This will be in the general form:

$$W_{it} = \beta X_{it} + \mathbb{C}A_{it} + v_i + \varepsilon_{it}$$

(The Addition of a mood coefficient (A_{it}) to general form model from Adler and Fleurbaey (2019, p. 428))

Where W_{it} is self-reported Well-being X_{it} is the sociodemographic factors we are measuring the impact of, A_{it} is the self-reported mood/affect, v_i are the unseen cognitive factors in individuals and ε_{it} includes other unobserved characteristics and measurement error (Adler and Fleurbaey, 2019, p.428).

In theory, this should account for differences in affective states. However, this comes with complications. Dr Barrett (2008, p. 3) stresses the importance of emotional granularity, which is an individual's ability to differentiate and accurately describe their emotions (p.106). Barrett states that emotional granularity is influenced by cultural and linguistic factors and that individuals don't have perfect emotional granularity but can improve their granularity through learning and practice (p.181). This implies that individuals can't fully and

accurately grasp their different affective states, as well as accurately describe them. This introduces measurement error in self-reported mood variable A_{it}, with the variable not reflecting true affective states, leading to biased estimates of the coefficient of A_{it}.

3.3 The use of Instrumental Variables

Another way to control for one's affective state is by using instrumental variables which can be used as a proxy for mood. For example, air quality (Cheng et.al, 2022) and weather conditions (Chhaochharia, 2019) have been used as proxies in economic literature, which are both examples of factors influencing one's affective state.

Chhaochharia et al. (2019, p.434) specifically used sky-cloud cover as a proxy for mood, by obtaining hourly sky-cloud cover data from US weather stations. More specifically, they used Desasonalised Sky Cloud Cover (DSKC) which provides a measure of weather patterns that could affect mood and decision-making, when accounting for normal seasonal variation (p. 434). The study correlates DSKC with decision-making in management settings, highlighting the influence that weather has on mood (p.436).

This can be applied to control for mood in a subjective well-being model with sociodemographic factors. It would be in the form:

$$W_{it} = \beta X_{it} + \alpha DSKC_{it} + v_i + \varepsilon_{it}$$

(The addition of the DSKC coefficient from Chhaochharia et al. (2019) to a general form model found in Adler and Fleurbaey (2019, p. 428))

Where W_{it} is self-reported Wellbeing, X_{it} is the sociodemographic factors we are measuring the impact of, $DSKC_{it}$ is the weather conditions being used as a proxy for affective state, v_i are the unseen cognitive factors in individuals and ε_{it} includes other unobserved characteristics and measurement error (Adler and Fleurbaey, 2019, p.428).

This is an example of a control mechanism that can be used to account for the time-varying nature of mood/affective state. Weather conditions are a useful tool because of their unlikeliness to cause endogeneity, as this variable is unlikely to be correlated with the error term of any economic model (Bound et al., 1995, p.443). Furthermore, it is unlikely to cause exclusion restriction violation, as weather conditions would only affect a person's affective state. However, it is important that the effect of weather on mood and subjective well-being may vary among individuals, the same weather conditions can have different effects on different people (Harley, 2019), and this makes it a weak instrument with inconsistent estimates.

4. Final Conclusion

4.1 Summary

This analysis explores the role of non-rational human psychology in the economics of happiness by applying the relatively new "theory of constructed emotions". This development in psychology has provided new insight and context to popular methods of modelling subjective well-being. Dr Barrett's work has provided more awareness of the nature of the data collected and how this affects assumptions made within models. Dr Barretts' insistence on the interplay between affect and cognition which for so long has been assumed not to be the case in Economics, creates new statistical challenges to be dealt with. The inference has been made about the possible ways to mitigate the time-invariant and time-variant heterogeneity of cognition and affect. When modelled on data from GSOEP, models with fixed effects had differing coefficients compared to those without (Ferrer-i-Carbonell and Frijters, 2004, p. 653). This demonstrates, along with other time-invariant factors, the implication of heterogeneous cognitive construction on subjective well-being models, which needs to be accounted for to attain more accurate coefficients for the effects of sociodemographic factors.

The time-varying nature of affect is a less-trodden path in the economics of happiness, as it has long been assumed that one could separate it from models (Adler and Fleurbaey, 2016, p. 425). The use of self-reported and proxy variables has been used but not fully explored.

4.2 Suggestions for Research

Using instrumental and self-reported variables for assessing for one's affective state/mood comes with many flaws, outlined above. More comprehensive ways of accurately measuring an individual's mood are needed. While Diener and Emmons's (1984) self-reported mood scale provides a good basis, Dr Barrett (2017, p. 16) insists that a one-dimension scale will not fully capture an individual's affective state. A possible combination of physiological measures, behavioural observations as well as self-reported questionnaires could form an affective variable with a high degree of accuracy. If we could use this regression analysis in subjective well-being models, we would be able to compare the results, and empirically understand the effect of affect.

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